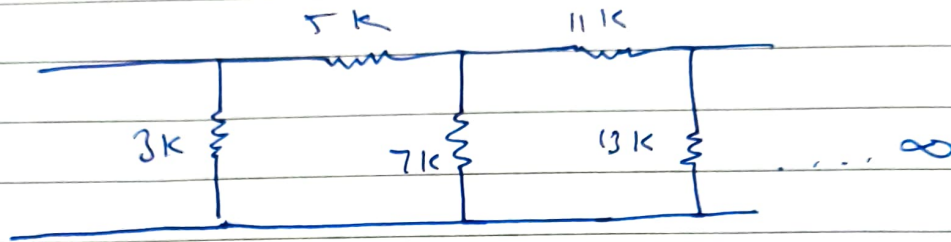


Circuits & Systems

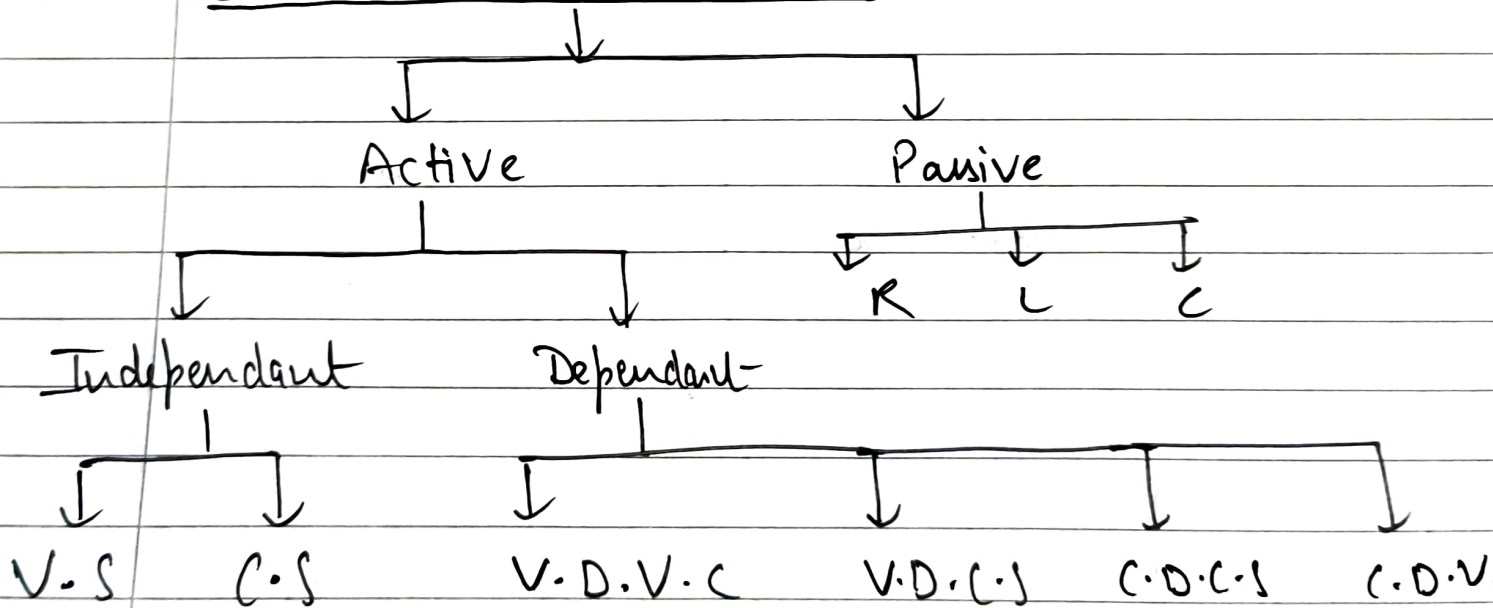
- Books -
- ① Alexander, M.O. Sadiku (Theory + Practice)
 - ② Van Valkenburg (Theory)
 - ③ A.K. Chakroborty (Numerical Practice)

Ques:-



Req. = ?

Basic Circuit Element



Active Element :-

- Deliver Energy ~ power
- Doesn't require any external force or energy source.
- * → Amplifier is also an Active Element.

Dependant Sources :-

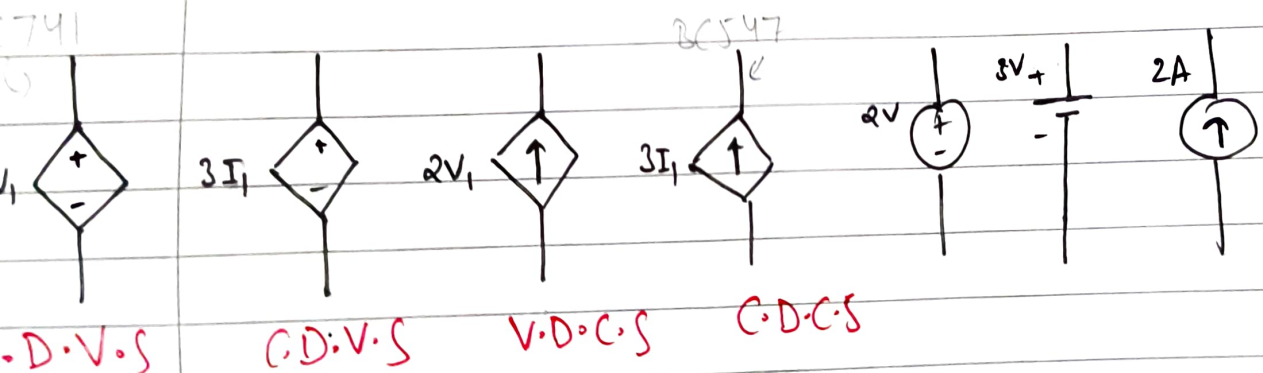
- Active Elements, whose value is depending on other parameter (it is not const.),

Eg → ① Voltage Dependant voltage source

② Voltage Dependant current source

③ Current Dependant voltage source

④ Current Dependant current source



Passive Elements

- ① R → Dissipates Heat
→ ohmic losses
→ ohm's law $V \propto I \Rightarrow V = IR$

- ② L → Stores Magnetic field
→ the current is stored
→ $V = \frac{d\psi}{dt} = \frac{d(N\phi)}{dt}$
 $= \frac{d(LI)}{dt} = L \frac{dI}{dt}$

$$\begin{cases} \psi = N\phi \\ \psi \propto i \\ \psi = Li \end{cases}$$

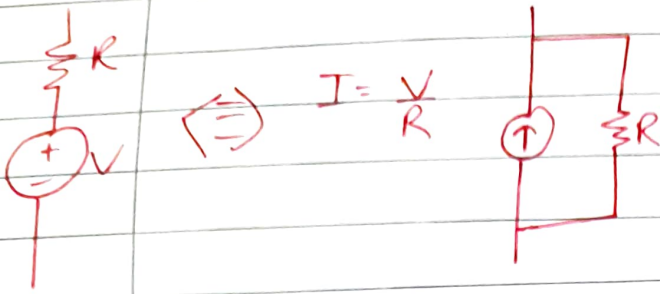
*** → Inductor doesn't allow sudden change in the current.

- ③ C → Electric field store karta hai
→ voltage store karta hai
→ $Q \propto V \Rightarrow Q = CV \rightarrow \frac{dQ}{dt}$
→ $\frac{dQ}{dt} = \frac{d(CV)}{dt}$

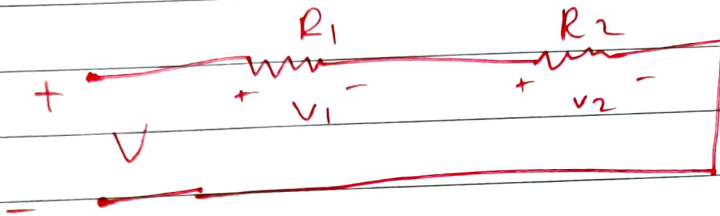
* Capacitor doesn't allow sudden change in the voltage.

Source Transformation

$I \rightarrow$ dc current
 $V \rightarrow$ dc voltage
 $i \rightarrow$ AC current
 $v \rightarrow$ AC voltage

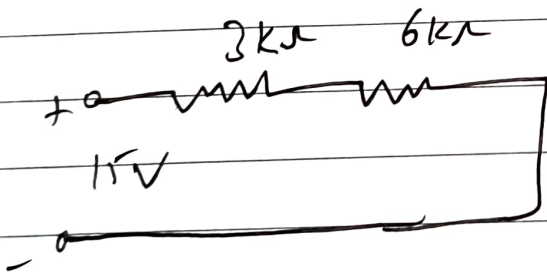


Voltage Division Rule



$$V_1 = \frac{V \times R_1}{R_1 + R_2} ; V_2 = \frac{V \times R_2}{R_1 + R_2}$$

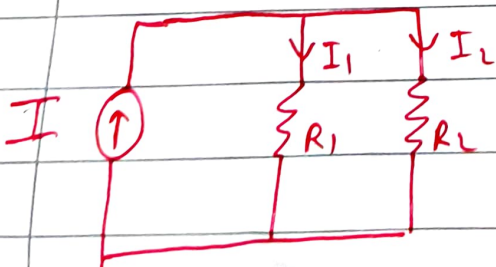
Ques $\rightarrow$



$$V_1 = \frac{15 \times 3}{9} = 5V$$

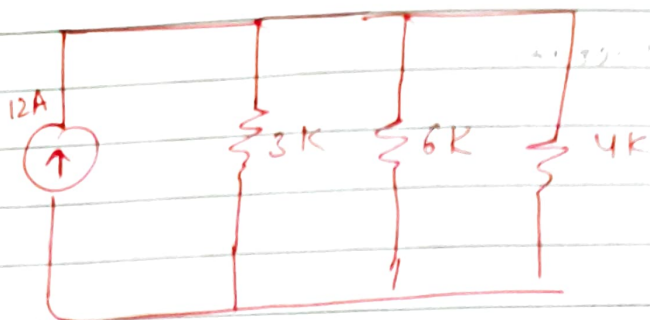
$$V_2 = \frac{15 \times 6}{9} = 10V$$

Current Division Rule



$$I_1 = \frac{I \times R_2}{R_1 + R_2} ; I_2 = \frac{I \times R_1}{R_1 + R_2}$$

Ques →



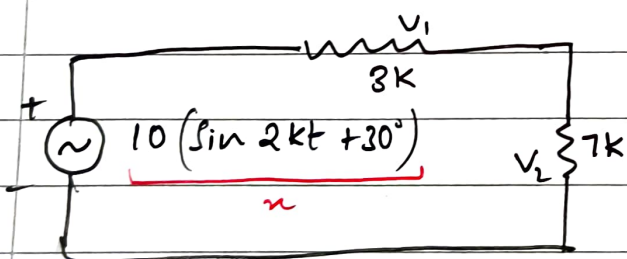
$$I_1 = \frac{12 \times \frac{1}{6}}{\frac{1}{3} + \frac{1}{6} + \frac{1}{4}} = \frac{12 \times \frac{2}{12}}{\frac{4}{12} + \frac{2}{12} + \frac{3}{12}} = \frac{12 \times \frac{2}{12}}{\frac{9}{12}} = \frac{12 \times 2}{9} = \frac{24}{9} = 2.66 \text{ A}$$

$$I_2 + I_3 = \frac{6 \times 4}{10} = 2.4$$

$$I_1 = \frac{I \times 2.4}{2.4 + 1} = \frac{12 \times 2.4}{5.4} = \boxed{5.33 \text{ A}}$$

$$I_2 = \frac{12 \times (12/7)}{6 + (12/7)} = \boxed{2.67 \text{ A}}$$

$$I_3 = \frac{12 \times (18/9)}{2 + 4} = \boxed{4 \text{ A}}$$



$$V_1 = \frac{V \times 3}{10} = \underline{0.3V}$$

$$V_2 = \frac{V \times 7}{10} = \underline{0.7V}$$

$$V_1 = 3.0 (\sin 2kt + 30^\circ)$$

$$V_2 = 7.0 (\sin 2kt + 30^\circ)$$

Superposition theorem

when a circuit consist of multiple active sources, then the response across any element, will be algebraic sum of individual responses generated by each active source acting alone.

Key features :-

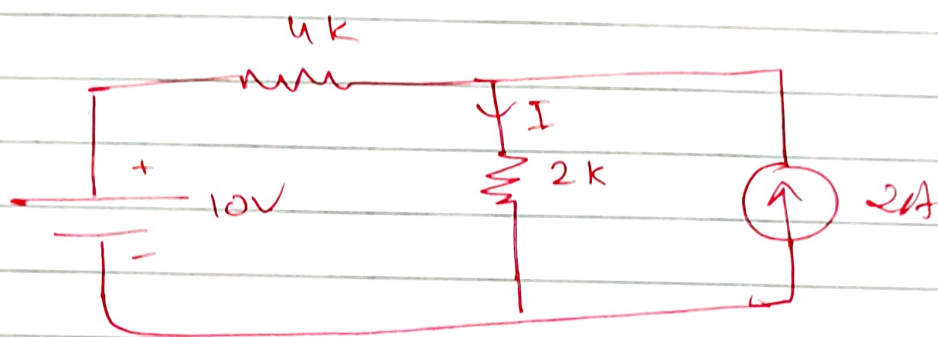
- ① ~~linear~~ Applicable for linear bidirectional / bilateral circuit.
- ② Complex Circuit $\rightarrow$ Easier for solⁿ purpose.

Limitations

- ① Doesn't work for non-linear circuit.
- ② Doesn't work for unidirectional / unilateral circuits.
- ③ Sometimes superposition theorem gives false result, in case of ideal circuits.

Bidirectional Elements

Ques → Calculate the value of current I using Superposition Theorem.



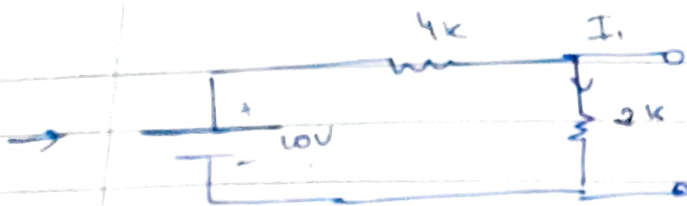
Case 1 →

when 10V source is Active

NOTE → When a signal source is Active, then remaining sources will be replaced by their internal resistance.

(1) Internal Resistance for Ideal Voltage Source = 0 (short circuit)

(2) Internal Resistance for Ideal Current Source = ∞ (open circuit)



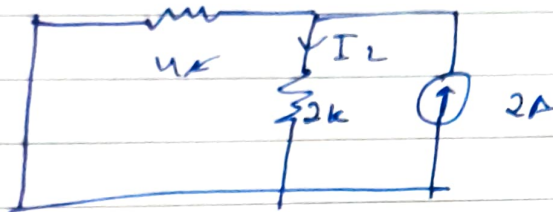
Case 1 →

$$V_{2k} = \frac{10 \times 2k}{6k}$$

$$= 10/3 \text{ V}$$

$$I_1 = \frac{10}{3} \text{ A}$$

Case 2 → when 2A source is active:



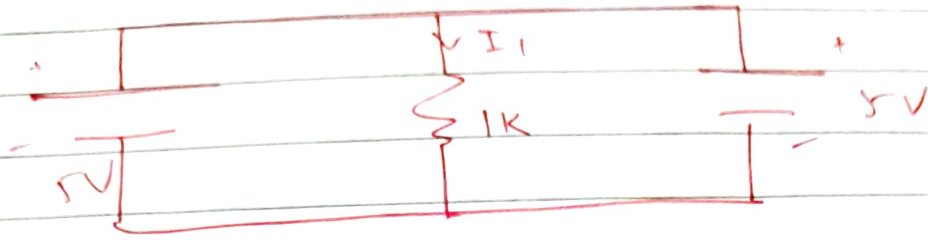
$$I_2 = \frac{2 \times 4}{6} = \frac{4}{3} \text{ A}$$

Step 3 → $I = I_1 + I_2$

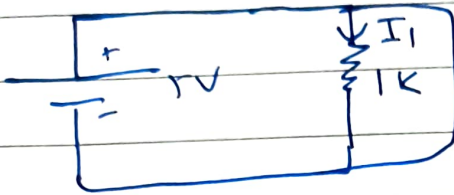
$$I = \frac{10}{3} + \frac{4}{3} = 3 \text{ A}$$

//_

Ques. Calc. the Value of Current I_1 using Super



Step 1 →



Thevenin's Theorem

- Application for linear networks
- " " bidirectional & bilateral ckt.

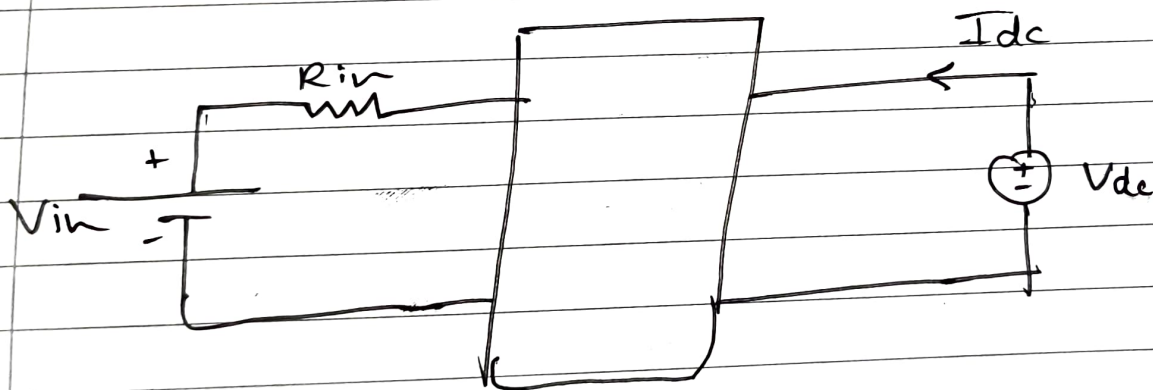
The process to apply thevenin theorem :-

step 1 → Remove the load (R_L or Z_L)

step 2 → Calc. V_{Th} across load

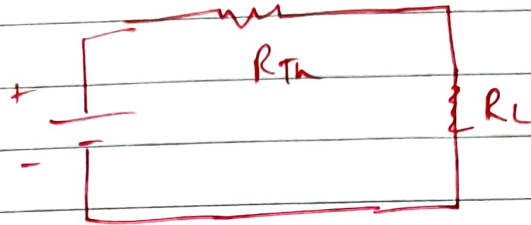
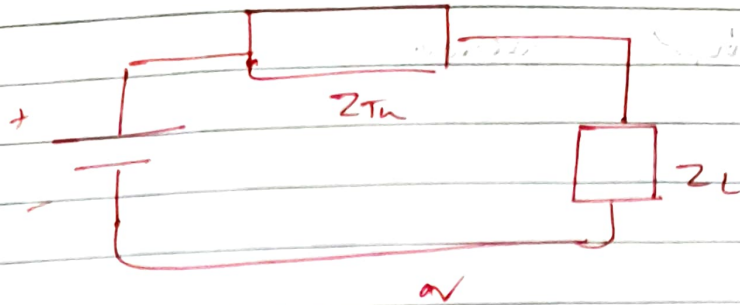
step 3 → Calc. R_{Th} or Z_{Th} across load

Note :- If any ckt consist of dependent sources then the process to calc. the average equivalence resistance is

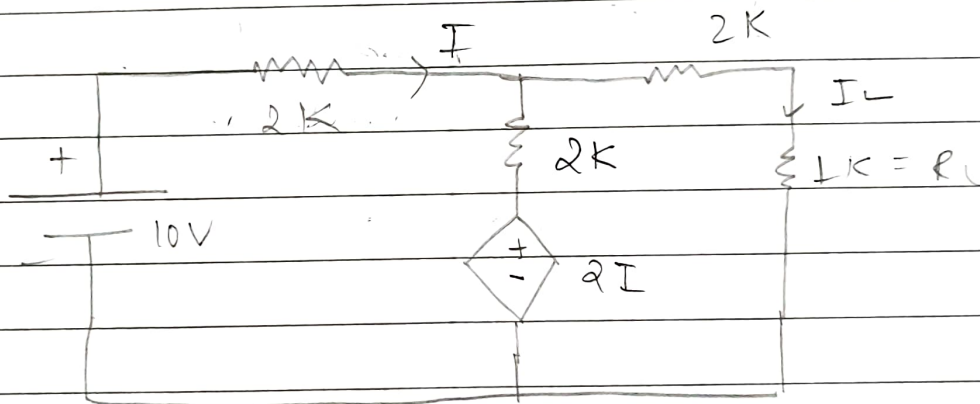


$$\frac{V_{dc}}{I_{dc}} = R_{Th} \quad \text{or} \quad \frac{V_{dc}}{I_{dc}} = Z_{Th}$$

Step 4 →

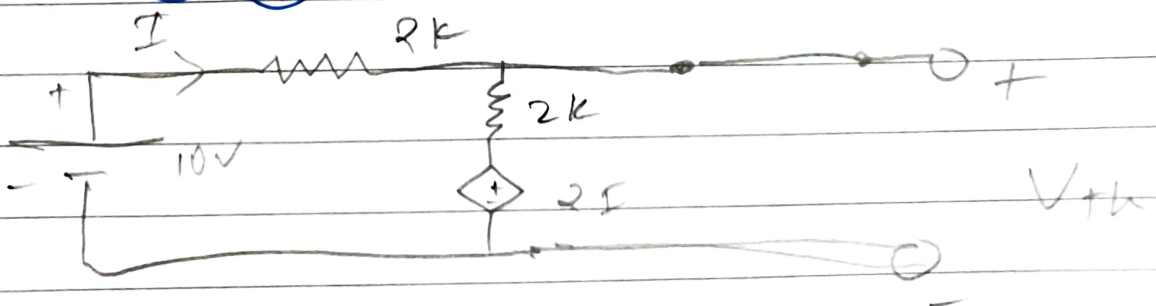


Ques → Determine the load current using Thevenin's theorem



$V = 0$	S.C
$R = 0$	
$I = 0$	O.C
$R = \infty$	

Step 1



Project-1

website $\rightarrow$ Tinker cad



Arduino

pin 3 $\rightarrow$ (+) LED^{red}

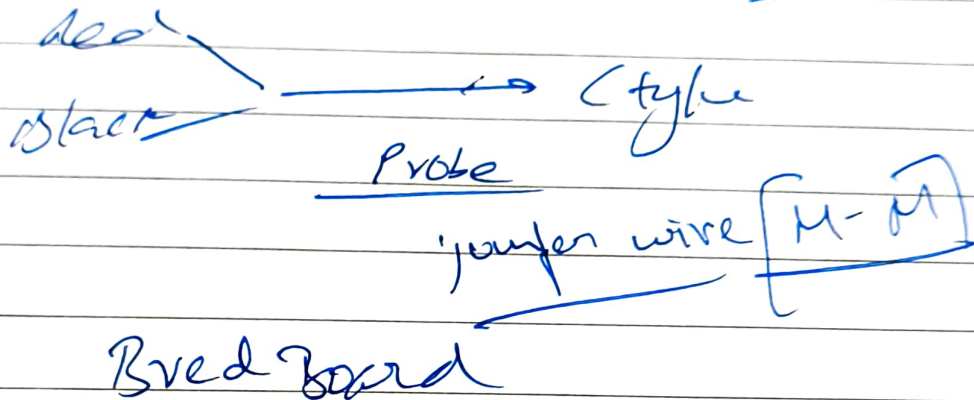
GND $\rightarrow$ (-)

Project-2

IN4007 $\rightarrow$ x4

10 μ F, 1 μ F $\rightarrow$ x1 each

7809 $\rightarrow$ 1



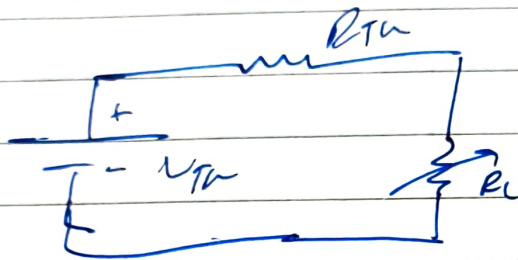
Maximum Power Transfer Theorem

Step 1 → Acc to this theorem, it is possible to calc. the max. power delivered to the load.

The process to calc. Max. Power is as follows:-

Step 2 → Apply THEVENIN'S Theorem

Step 3 →



$$I_L = \frac{V_{th}}{(R_{th} + R_L)}$$

$$P_L = \frac{V_{th}^2}{(R_{th} + R_L)}$$

$$\frac{dP_L}{dR_L} = V_{th}^2 \frac{d}{dR_L} \left(\frac{1}{R_{th} + R_L} \right)$$

$$= V_{th}^2 \left[-\frac{1}{(R_{th} + R_L)^2} \right]$$

****** $P_L = I_L^2 R_L \rightarrow \frac{V_{th}^2}{(R_{th} + R_L)^2} \times R_L$

$$\frac{dP_L}{dR_L} = V_{th}^2 \frac{d}{dR_L} \left[\frac{R_L}{(R_{th} + R_L)^2} \right]$$

$$\rightarrow \frac{V_{th}^2 \left[(R_{th} + R_L)^2 - R_L(2)(R_{th} + R_L) \right]}{(R_{th} + R_L)^4} = 0$$

$$(R_H + R_L)^2 - 2R_L(R_H + R_L) = 0$$

$$R_H^2 + R_L^2 + 2R_H R_L - 2R_H R_L - 2R_L^2 = 0$$

$$R_H^2 - R_L^2 = 0$$

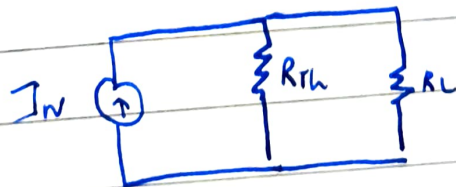
$$R_H^2 = R_L^2$$

Norton's Theorem

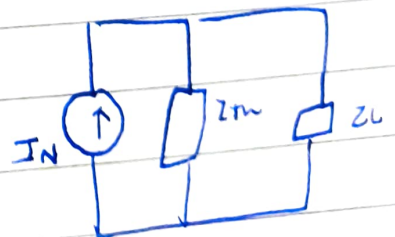
- ① Applicable for linear crkts.
- ② Valid for bidirectional/ bi-lateral.
- ③ Analogous for Thevenin's theorem.
- ④ The procedure to apply Norton's theorem is:

- Step } i) Remove the load R_L or Z_L
- ii) Calc. of Equivalent resistance or impedance across the load.
[exactly similar to Thevenin equivalent resistance
or Thevenin eqn- impedance.]
- iii) Replace load by short crkt- & calc. short ckt current (I_N)

iv) Norton's Equivalent- Model



or



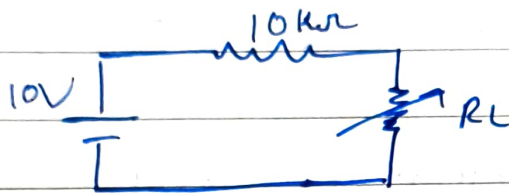
Maximum Power for Impedance Z_{th}

$$\text{Condition} = Z_L = Z_{th}^*$$

* = complex conjugate ($i = -i$)

Ques 1 for what value of load resistance, max power will be delivered

Ques 2. Determine Total Power delivered by source



$$\text{A1)} \quad = R_L = R_{th} \Rightarrow 10 \text{ k}\Omega$$

$$\begin{aligned} 2) \quad P &= V \times I \\ &= \frac{10}{20} = 0.5 \text{ mW} \end{aligned}$$

$$\underline{I \times V = 0.5 \text{ mW}}$$

Q2i) what is the relation b/w max. power consumed by load & total power supplied by source.

Q7) Determine the power consumed by R_m

Q8. Efficiency of the network = $\eta = 50\%$

$$\frac{\text{Power Delivered to load } P_{LW}}{\text{Total power supplied}} = \frac{0.25}{0.5} = 1W = 50\%$$

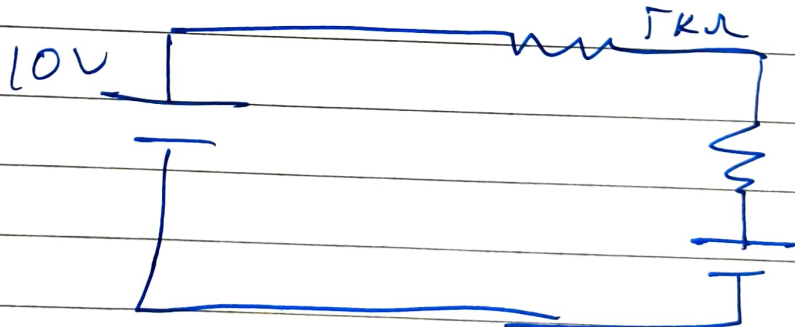
Q7) Voltage across load = 5V

Q8) Relation b/w voltage across load & total voltage by source.

NOTE → Resistance Network

$$P_{\text{max}} \rightarrow V_s \rightarrow V_s/2$$

$$P_s = P_s/2$$



Determine the value of R for which max. power is delivered to the load

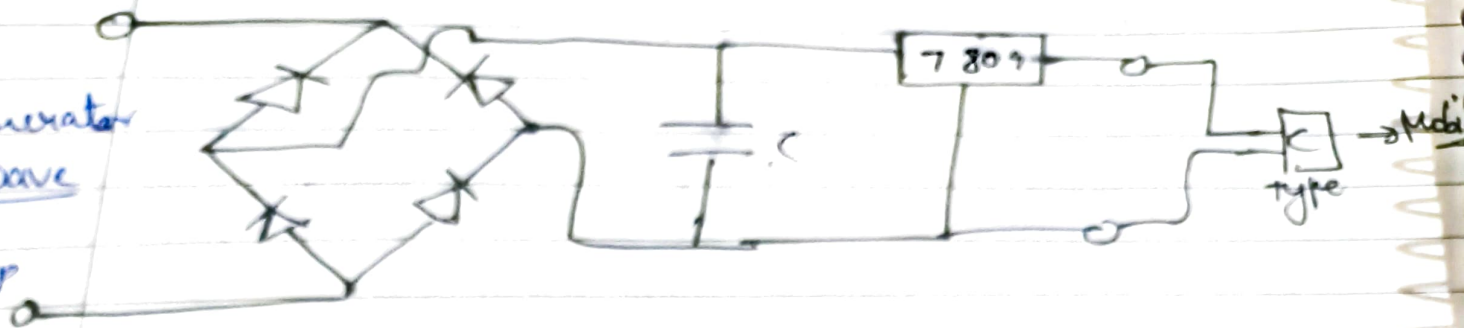
→ [what is Max power]

→ [what is efficiency]

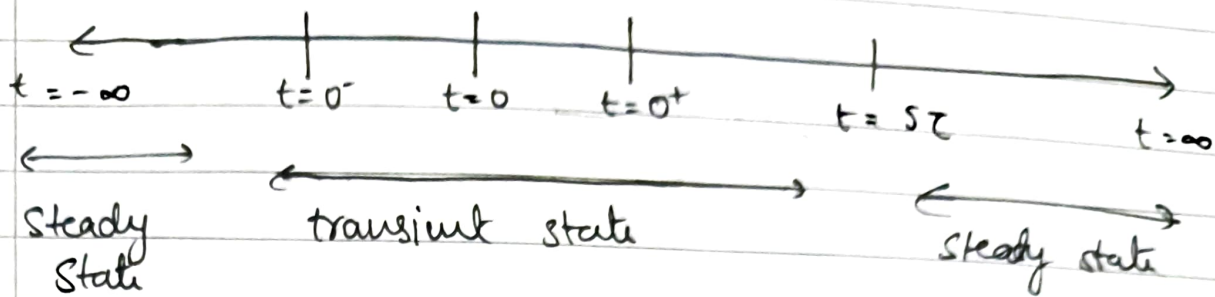
fun. generator
Sin wave

20Vpp

freq. 50Hz



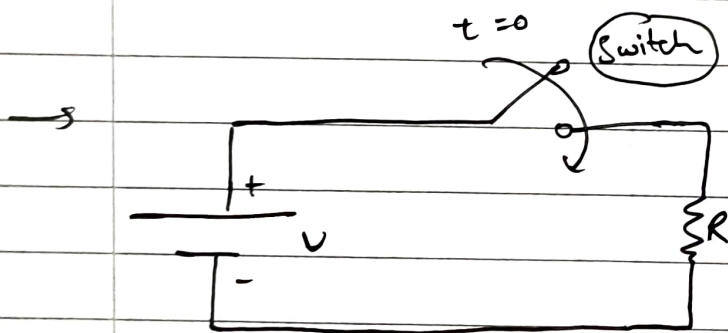
Transient analysis



$t = 0 \rightarrow$ present time

$t = 0^- \rightarrow$ immediate before the operation of switch
(just before $t = 0$)

$t = 0^+ \rightarrow$ after the operation of switch
(just after $t = 0$)



Step 1 $\rightarrow$ at $t = 0^-$

Switch = open
"steady state"



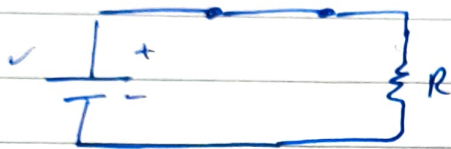
$$I(0^-) = 0$$

Step 2 $\rightarrow$ At $t = 0^+$ Switch = close
transient state



$$I(0^+) = \frac{V}{R}$$

Step 3 $\rightarrow$ At $t = \infty^+$ switch close
steady state

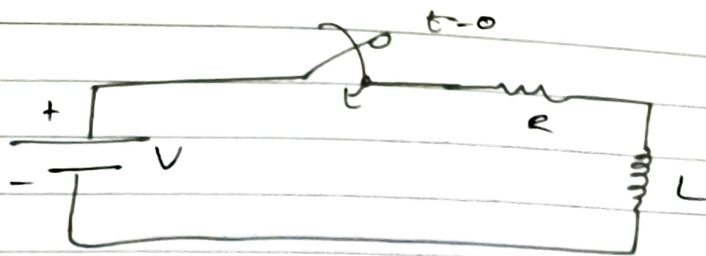


$$I(\infty^+) = \frac{V}{R}$$

Note $\rightarrow$

Resistor Allows sudden change in the voltage & current.

Leves



Step 1 $\rightarrow t = 0^-$

SW = open "steady state"

$$I_R(0^-) = 0 ; I_L(0^-) = 0$$



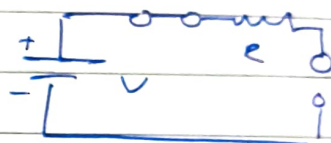
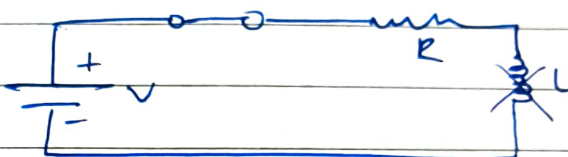
~~Step 2~~

Step 2 $\rightarrow$

$t = 0^+$

SW = close

"transient state"



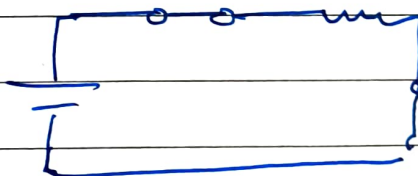
$$I_R(0^+) = 0 ; I_L(0^+) = 0$$

Step 3 $\rightarrow$

$t = \infty$

SW = close

"steady state"



Short ckt

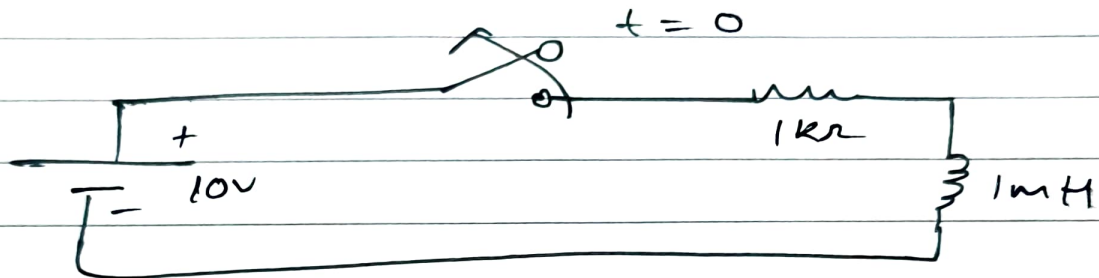
$$(\infty) I = \frac{V}{R}$$

$$x(t) = \left[x(0^+) - x(\infty) \right] e^{-(t/\tau)} + x(\infty)$$

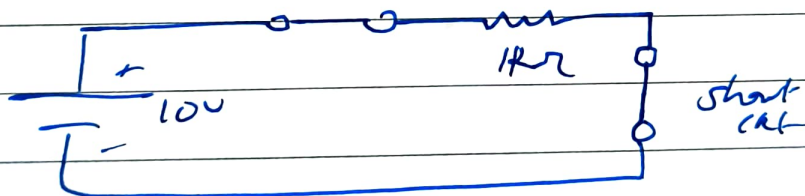
τ = time constant of ckt

$$\begin{aligned} \tau &= \text{time const} \\ \tau &= RC \\ \tau &= \frac{L}{R} \end{aligned}$$

Ques



Step 1 $\rightarrow$ At $t = 0^-$
sw = close "Steady state"



$$I_L(0^-) = 10 \text{ mA}$$

Step 2 → At $t = 0^+$

sw = open

"Transient state"



$$I_R(0^+) = 0$$

$$I_L(0^+) = 10 \text{ mA}$$